

1. Software Architecture & CI/CD Release Protocols

ApexTech follows a cloud-native microservices architecture communicating via gRPC for high-throughput internal RPCs and REST OpenAPI v3 for external integrations.

Code Review & Quality Thresholds:

- Every Pull Request requires a minimum of 2 peer approvals before merging into the main branch.
- Automated CI/CD pipelines require a minimum unit test coverage threshold of 85% and zero high-severity SAST vulnerabilities.

Release Windows:

- Production deployments occur exclusively during designated Release Windows: Tuesdays and Thursdays between 10:00 AM and 2:00 PM UTC.
- Emergency hotfixes require explicit sign-off from the VP of Engineering and the On-Call Incident Commander.

2. Cloud Infrastructure, Security & Compliance

Data Encryption Standards:

- All customer data at rest must be encrypted using AES-256 bit encryption key standards via Cloud KMS.
- Data in transit is strictly enforced with TLS 1.3 protocol encryption.

Key Rotation & Access Governance:

- Master encryption keys in Cloud KMS are automatically rotated every 90 days.
- Production environment access is granted strictly via Zero-Trust Just-In-Time (JIT) access approval lasting a maximum of 4 hours per session.

Data Retention & Compliance:

- ApexTech is certified SOC 2 Type II, ISO 27001, and GDPR compliant.
- Inactive user logs and telemetry data are automatically purged after 365 days of inactivity.

3. Machine Learning & RAG Search Architecture

Retrieval-Augmented Generation (RAG) Architecture:

- Text Chunking Strategy: Document parsing utilizes a recursive character splitter with a target chunk size of 500 tokens and a 100-token overlap.
- Embedding Model: Dense vector representations are computed using the SentenceTransformers 'all-MiniLM-L6-v2' model with a 384-dimensional vector space.
- Distance Metric: Vector similarity searches utilize Cosine Similarity distance metrics within ChromaDB vector stores.

LLM Fallback & Performance Targets:

- Primary LLM: Google Gemini 1.5 Flash (optimized for low-latency generation).
- Secondary Fallback: Google Gemini 1.5 Pro (triggered if context length exceeds 32k tokens).
- Latency Target: Vector retrieval must complete within < 120ms, and total end-to-end user query response time must remain under 1.8 seconds.

4. Employee Compensation, Wellness & Perks

Health & Wellness Benefits:

- Health Insurance: 100% premium coverage for full-time employees and 80% coverage for eligible family dependents.
- Annual Wellness Stipend: Every employee receives a \$1,500 annual flexible stipend usable for gym memberships, sports gear, or mental health applications.

Sabbatical & Long Service Rewards:

- Paid Sabbatical: Employees completing 4 consecutive years of full-time service are eligible for a 4-week fully paid sabbatical leave.
- Relocation Assistance: Up to \$8,000 relocation reimbursement is offered for approved domestic or international office transfers.

5. Disaster Recovery (DR) & Incident Response SLAs

Disaster Recovery Metrics:

- Recovery Time Objective (RTO): Target RTO is 15 minutes for critical database clusters during regional failovers.
- Recovery Point Objective (RPO): Target RPO is 5 minutes maximum allowable data loss for transactional stores.

Incident Classification & Response SLAs:

- Severity 1 (Sev-1): Critical production outage affecting > 25% of active users. SLA Response: On-call engineer must acknowledge within 10 minutes and provide status updates every 30 minutes.
- Severity 2 (Sev-2): Non-critical feature degradation. SLA Response: Acknowledged within 30 minutes, update every 2 hours.
- Severity 3 (Sev-3): Minor bug with workaround. SLA Response: Addressed in regular sprint cycle within 5 business days.